

► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

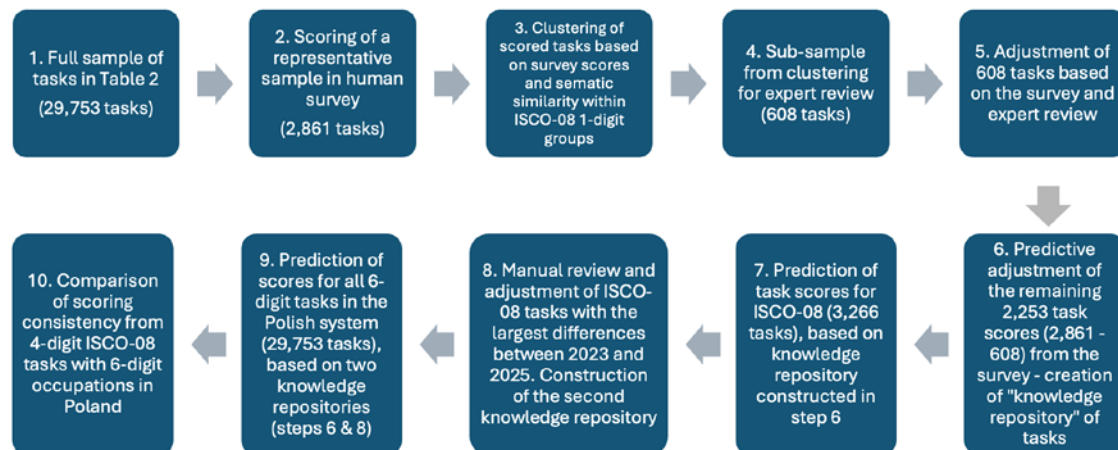
Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people

► 4 Expert validation survey

The expert group includes international specialists at the ILO and its partner organizations, NASK and the Polish Ministry of Family, Labour and Social Policy.⁹ The primary criterion for expert selection is their understanding of labour markets, including the complexity of occupations and tasks, and their real-life applications. While the experts are expected to have familiarity with the current family of generative AI technologies and their applications, expertise in AI design or development is not a requirement. This approach ensures that the focus remains on the feasibility and practical implications of automation within existing technological constraints, as technology experts tend to overstate technological capacities and risks when evaluating broader applications, particularly in workplace settings (Karger et al., 2023). This tendency is evident in earlier automation studies, which often assigned high displacement potential to routine tasks or entire occupations, such as garment production. As de Mattos et al. (2020) illustrate, these technological predictions frequently failed to account for the challenges of automating tasks requiring high dexterity or handling complex materials, such as stitching pliable fabrics. Accordingly, the assessment of automation potential in our study prioritizes expertise of task execution and workplace realities over purely technological expertise.

Since the experts cannot review all 2,861 tasks used in the survey, we focus on a representative sub-sample, selected in such a way to subsequently permit adjustments to entire task clusters. The overall scheme of this score adjustment procedure is presented in Figure 11.

► Figure 11. Schema of the scoring stages



First, we cluster the tasks into four fixed intervals, based on the distribution of the mean scores from the survey.¹⁰ We then use hierarchical clustering to group similar tasks within those clusters based on their semantic content. First, we assign embeddings to each task description using the *text-embedding-ada-002* model from OpenAI. Next, for each subgroup – defined by the combination of an occupational group at ISCO-08 1-digit level and a particular survey score cluster – we

⁹ A full list of contributors is provided in the acknowledgements section.

¹⁰ $\mu \in [0, 0.1] \cup (0.1, 0.3] \cup (0.3, 0.5] \cup (0.5, 1]$

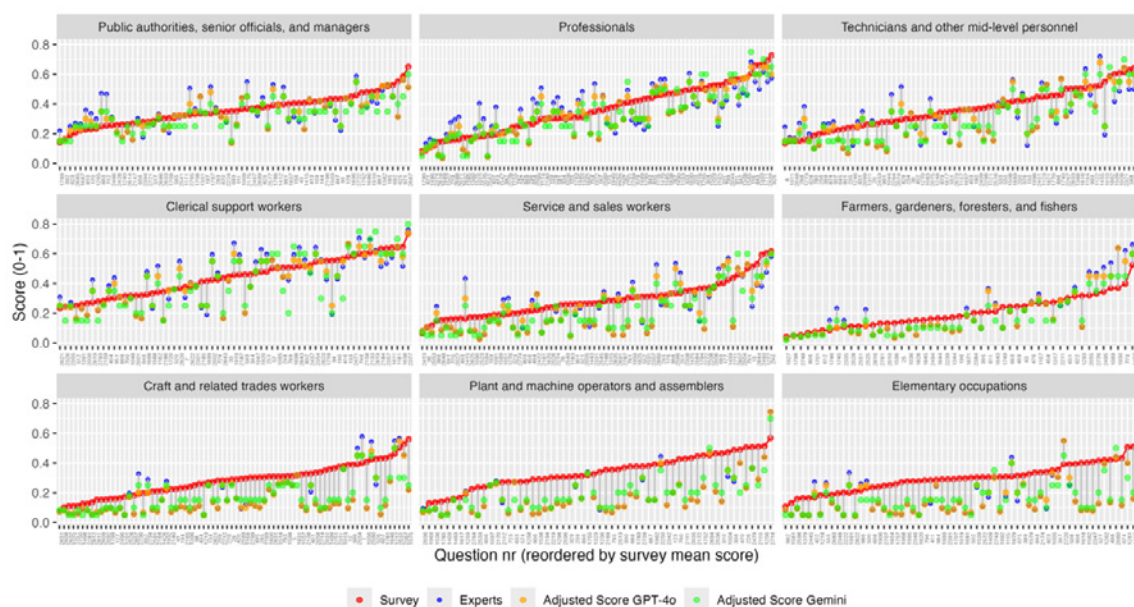
Each of the selected 608 tasks was then presented to a panel of experts. For each task, the experts were shown (i) the occupation and cluster information, (ii) the average survey score (rescaled to 0–1), and (iii) the full task description. If they agreed with the survey score, they left it unchanged; otherwise, they entered a revised expert score (0–1).

In many cases, the differences between the survey and expert scores are non-trivial. They also vary significantly across occupational groups, revealing distinct patterns based on the nature of pertaining tasks (Figure 13). In occupations dominated by manual and physical tasks, such as Craft and Related Trades Workers, Plant and Machine Operators, and Farmers, Gardeners, Foresters, and Fishers, experts consistently assigned lower scores compared to the survey respondents, reflecting the practical limitations of automating tasks that require physical manipulation or manual dexterity. In contrast, occupations involving more cognitive or knowledge-intensive tasks, such as Clerical Support Workers and Professionals, show a much more mixed picture. While some tasks in these groups received lower scores, many others were revised upwards by the expert group, reflecting the greater adaptability of generative AI for structured, data-driven, or communication-focused tasks. For example, Clerical Support Workers display a combination of increases and decreases, suggesting nuanced task-specific assessments where automation potential depends heavily on the nature of individual tasks.

► 5 Adjustment of survey sub-sample

In the next step, we compared the average of expert scores with the original mean survey scores using two independent AI models as impartial arbiters to propose an adjusted score.¹³ The models were informed about the sources of the scores in such a way that would prevent them from automatically siding with the expert group: group one was defined as *“practical knowledge of the work and types of tasks they were presented, since they worked in the same occupational groups from which we selected tasks they scored”*, while the second group was described as *“respondents with more theoretical knowledge of technology and labour markets”*. Rather than simply averaging the two scores, the models were explicitly instructed to focus their analysis on the task characteristics and the current capabilities of generative AI and required to provide justifications for each adjustment and its relative position to the original scores.

► Figure 13. Task-level scores from the survey and experts, compared to AI-arbitrated scores

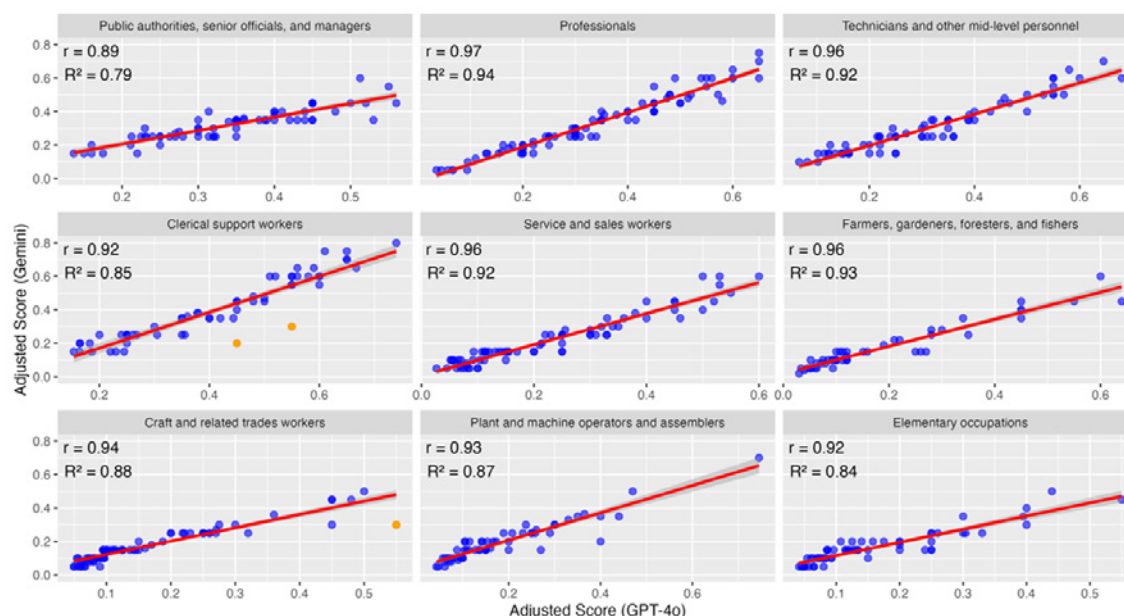


The overall correlation between the adjusted scores produced by the two AI models is 0.96, with correlations within occupational groups exceeding 0.9 in all but one case¹⁴ (Figure 14). This high level of agreement demonstrates that, when provided with sufficient background knowledge in the form of reference ranges of scores, the models converge on remarkably similar predictions. Notably, only three tasks exhibited an absolute difference in scores exceeding 0.2 (orange dots in Figure 13). In these cases, GPT-4o tended to adopt a more optimistic perspective on the feasibility of automation, whereas Gemini emphasized the importance of human judgment as a critical factor.

¹³ We used OpenAI's GPT-4o and Google's Gemini Flash 1.5, which have comparable abilities. We also tested OpenAI o1, which at the time of writing was among the models with strongest capacities of « algorithmic reasoning » and observed very similar results. The process was set up as a cycle of sequential API calls, which loop over each task and capture the response into a data frame.

¹⁴ Public authorities, senior officials and managers in Figure 10.

► Figure 14. Comparison of Adjusted Scores: GPT-4o vs Gemini by Occupational Group



For the final adjustment, we averaged the scores of the two arbitrating models and manually reviewed the justifications, without observing any significant inconsistencies. Table 4 illustrates the result of the adjustment process through examples of tasks where expert scores were significantly higher or lower than the survey averages, with the final adjusted score and GPT-4o justification. For tasks such as “operating injection molding machines” or “performing mechanical processing of silicon”, experts highlighted the limitations of generative AI in handling physical and manual components, leading to significantly reduced adjusted scores compared to the survey. Conversely, for tasks like “maintaining documentation of classes with children” or “performing financial estimates for household budgets”, the experts recognized the structured and analytical nature of these tasks, which generative AI can assist with to a substantial degree.

► Table 4. Sample of adjustments and justifications between the survey and expert scores (examples of largest upward and downward revisions)

Task				Justification
Applying various welding methods and using different types of welding machines and equipment, in accordance with training or qualifications.	0.43	0.05	0.05	The task involves hands-on physical work requiring skill and precision, such as welding using various machines, which is not currently automatable by generative AI technologies like ChatGPT. These technologies excel in tasks involving data processing, language understanding, and information generation but do not possess capabilities for physical task execution, hence aligning closer with the revised score.